Junsup Shim | Curriculum Vitae

Dept. of Earth Science Education, Pusan National University
2, Busandaehak-ro 63 beon-gil, Geumjeong-gu, Busan, Korea 46241
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Academic Positions

Assistant Professor *Mar 2026 – present*
Pusan National University, Korea

Distinguished Postdoctoral Fellow *Jul 2022 – Feb 2026*
Academia Sinica Institute of Astronomy and Astrophysics, Taiwan

Visiting Scientist *Oct 2023 – Oct 2023*
Korea Institute for Advanced Study, Korea

Visiting Scientist *Jul 2023 – Aug 2023*
Canadian Institute for Theoretical Astrophysics, Canada

Research Fellow *Sep 2018 – Jun 2022*
Korea Institute for Advanced Study, Korea

Education & Degrees

Ph.D. in Astronomy, Seoul National University, Korea *Aug 2018*
Advisor: Prof. Jounghun Lee
Thesis: Cosmological Applications of Filamentary Structures in the Universe

B.S. in Mechanical and Aerospace Engineering, Seoul National University, Korea *Aug 2011*
Cum laude

Publication Summary (Source: NASA Astrophysics Data System)

Total Publications: 11 papers in peer-reviewed journals with 119 citations

- **First Author Publications:** 10 papers with 94 citations
- **Coauthor Publication:** 1 paper with 25 citations

Manuscripts Under Review: 2 articles

Unpublished Manuscripts: 2 articles (on arXiv) with 15 citations

Research Excellence

NRF Early-Career Research Grant *2026 – 2031*
Funded as PI (150 million KRW/yr) by National Research Foundation of Korea
Project title: "Probing Cosmic Parity Symmetry using the Large-scale Structure of the Universe"

PRL paper featured as Synopsis in Physics Magazine *2025*
Synopsis title: *"Cosmic Handedness Might Show Up in Galaxy Spins"*
Publisher: American Physical Society (Physics 18, s126)

Publications in Premier Journals *2025*
First author papers in PRL (IF: 9.0) and ApJL (IF: 11.7)

National Junior Research Fellowship *2014 – 2018*
Funded as PI (44,220 USD/yr) by National Research Foundation of Korea
Project title: "Identifying the Ultimate Cosmology by utilizing Supercluster Straightness"

Research Interests

– Parity symmetry of the Universe
– Gravity on cosmological scales
– Nature of dark energy and dark matter
– Large-scale structure of the Universe

Talks at International Meetings, Colloquia, and Seminars in 3 years

22. **[Conference, invited]** Kavli Institute for the Physics and Mathematics of the Universe, Japan, Oct 25, 2026, "*Which filaments matter: The relative scales of anisotropic infall*"
21. **[Summer Institute, invited]** Korea Institute for Advanced Study, Korea, Jul 14, 2026, "*Probing Cosmic Parity Symmetry with Large-Scale Structure of the Universe*"
20. **[Workshop, invited]** Asia Pacific Center for Theoretical Physics, Korea, Jun 24, 2026, "*Probing Cosmic Parity Symmetry with Large-Scale Structure of the Universe*"
19. **[Seminar, invited]** Institute of Physics at National Yang Ming Chiao Tung University, Taiwan, Dec 3, 2025, "*Probing Cosmic Parity Symmetry with Large-Scale Structure of the Universe*"
18. **[Conference]** Reconstructing the Universe, Taiwan, Dec 15 - 18, 2025, "*Improving cosmological constraints using galaxy intrinsic alignment in full-shape analysis*"
17. **[Conference]** The 3rd Shanghai Assembly on Cosmology and Structure Formation, China, Nov 16 - 21, 2025: "*Tightening Dynamical Dark Energy Constraints with Intrinsic Galaxy Alignments*"
16. **[Colloquium, invited]** Institute of Astronomy at National Central University, Taiwan, Oct 17, 2025: "*Tracing the Universe's Handedness Through Galaxy Spins*"
15. **[Colloquium, invited]** Institute of Astronomy at National Tsing Hua University, Taiwan, Oct 3, 2025: "*The Handed Cosmos: What Galaxies Reveal About Symmetry Breaking*"
14. **[Seminar]** Department of Physics at University of Seoul, Korea, Sep 24, 2025: "*The Handed Universe: What Galaxy Spins Can Tell Us*"
13. **[Conference]** CosKASI Conference 2025, Korea Astronomy and Space Science Institute, Korea, Sept 15 - 19, 2025: "*Tightening Dynamical Dark Energy Constraints with Full-shape Intrinsic Alignment of Galaxies*"
12. **[Conference]** International Symposium on Cosmology and Particle Astrophysics CosPA 2025, Institute for Basic Science, Korea, Jul 7 - 11, 2025: "*Tightening Dark Energy Constraints with Intrinsic Galaxy Shape Alignments*"
11. **[Conference, invited]** Parity Violation from Home (online conference), Nov 19 - 22, 2024: "*Probing Parity Violation in the Early Universe using Galaxy Spins*"
10. **[Colloquium]** Department of Astronomy at Yonsei University, Korea, Nov 8, 2024: "*Probing Parity Violation in the Early Universe using Galaxy Spins*"
9. **[Colloquium]** Korea Astronomy and Space Science Institute, Korea, Nov 6, 2024: "*Probing Parity Violation in the Early Universe using Galaxy Spins*"
8. **[Seminar]** Department of Physics and Astronomy at Seoul National University, Korea, Nov 5, 2024: "*Probing Parity Violation using Galaxy Spins*"
7. **[Workshop]** 11th KIAS Workshop on Cosmology and Structure Formation, Korea Institute for Advanced Study, Korea, Oct 27 - Nov 2, 2024: "*Probing Parity Violation in the Early Universe using Galaxy Spins*"
6. **[Conference]** COSMO'24, Kyoto University, Japan, Oct 21 – 25, 2024: "*Probing Parity Violation in the Early Universe using Galaxy Spins*"
5. **[Conference]** International Symposium on Cosmology and Particle Astrophysics CosPA 2024, Ningbo University, China, Jun 14 – 18, 2024: "*Probing Vectorial Parity Violation in the Early Universe*"

4. [**Workshop**] Large-scale Parity Violation Workshop, Academia Sinica Institute of Astronomy and Astrophysics, Taiwan, Dec 4 – 7, 2023: "*Detectability of Helical Vector Fossils in Galaxy Spin Maps*"
3. [**Seminar, invited**] Academia Sinica Institute of Physics, Taiwan, Jul 31, 2023: "*Cosmic Voids as Counterparts of Massive Clusters*"
2. [**Seminar, invited**] Canadian Institute for Theoretical Astrophysics, Canada, Jul 31, 2023: "*Cosmic Voids: Cluster Counterparts?*"
1. [**Seminar, invited**] Leung Center for Cosmology and Particle Astrophysics, Taiwan, May 3, 2023: "*Cosmic Voids: Cluster Counterparts?*"

Professional Service

- Reviewer, Journal of Cosmology and Astroparticle Physics, Nov 2025 – present
- Reviewer, Astrophysics and Space Science, Aug 2023 – present
- SOC, Parity Violation From Home, Nov 2025
- ASIAA Colloquium Committee, Aug 2022 – Oct 2024
- ASIAA Lunch Talk Organizing Committee, Aug 2022 – Oct 2024
- Journal Club Organizer, KIAS Astrophysics Group, Mar 2021 – Aug 2021
- Organizing Committee, 2020 Korea Astrophysics Workshop, Nov 2020
- Organizing Committee, 9th KIAS workshop on Cosmology and Structure formation, Nov 2020
- Organizing Committee, Survey Science Group Workshop, Feb 2020
- Organizing Committee, 2019 KIAS Astrophysics Group Workshop, Nov 2019

Mentoring Experience

- Master & Undergraduate Student Supervision

  – **Byoungjo Yu** (Master Y2) – Chungnam National Univ., Korea *Aug 2024 – present*
  Co-supervision with Prof. Hyunmi Song
  Project title: Boosting the sensitivity to primordial parity violation with optimally selected halos

  – **Tz-En Gao** (Undergraduate Y4) – National Taiwan Univ., Taiwan *Mar 2026 – present*
  Project title: Helicities of galaxies in the parity-violating universe

  – **Yu-Chien Hung** (Undergraduate Y3) – National Taiwan Univ., Taiwan *Jul 2025 – Dec 2025*
  Project title: Probing cosmic alignments: Investigating shape correlations beyond galaxies

Teaching Experience

- (**Invited**) Lecturer at ASIAA Summer School, "*Intro. to Modern Cosmology*", 2024 & 2025
- Teaching Assistant at Seoul Nat'l. Univ., "*Intro. to Astrophysics 2*", Spring 2014
- Teaching Assistant at Seoul Nat'l. Univ., "*Intro. to Modern Cosmology*", Fall 2013
- Teaching Assistant at Seoul Nat'l. Univ., "*Man and the Universe*", Spring 2012 & 2013

Language Proficiency

- Korean: Native
- English: Fluent